# Track 1 pipeline report

Team fulopjoz. MVA Hackathon 2026, Track 1, variant prioritization. Submission file: `submission_track1.csv`.

## Result

The pipeline ranks the clinically confirmed genotype first of 8,666 candidates: compound-heterozygous **BUB1B p.Leu737Ter** (chr15:40,209,701 T>G, stop gained; ClinVar allele ID 529272) plus **BUB1B p.Asn1002Lys** (chr15:40,220,612 T>G, missense), score 0.5823. Score at rank 2 is 0.5243. The score column `epcr` — estimated probability of causal relationship, the challenge-defined (0, 1] score — equals the pipeline ranking score.

| Rank | Genotype | Score |
|---|---|---|
| 1 | BUB1B p.Leu737Ter + p.Asn1002Lys | 0.5823 |
| 2-7 | RAI1 frameshift-deletion pairs (two orientations) | 0.5243 / 0.4914 |
| 8-10 | KIF7 pairs (founder allele is a common stop polymorphism, gnomAD frequency 0.561) | 0.4515 |
| 2,373 and below | homozygous-recessive singles (ordered after biallelic pairs by design) | — |

## Method

One pass over the 5,012,204-record proband VCF. Each stage's counter is measured from the run.

1. **Quality filter.** FILTER is PASS or absent, genotype quality at least 20, read depth at least 10, alternate-allele read fraction at least 0.25. Keeps 4,592,140.
2. **Effect calling.** Variants are annotated against local GENCODE v47 MANE transcripts. Stop gained, frameshift, and splice-site changes count as loss of function (LoF); missense and in-frame indels are scored separately. Keeps 16,963 coding or splice candidates.
3. **Rarity.** Allele frequency from local gnomAD v4.1 exome sites; every record at a multi-record position is merged and alleles match exactly on (chrom, position, ref, alt). ClinVar joins as background evidence only.
4. **Candidate pairing.** Compound-heterozygous pairs (an LoF allele plus a second damaging allele in the same gene; phase not required), rare homozygous missense, rare homozygous stop gain, rare homozygous in-frame indels. Finds 2,372 pairs and 6,294 homozygous candidates.
5. **Scoring.** Each variant scores as a weighted mix: founder effect 0.45, predicted pathogenicity 0.60, rarity 0.30, plus phenotype similarity to the eight case HPO terms, weight 2.0. A pair scores as the stronger allele $\times$ (0.5 + 0.5 $\times$ the weaker).

## External cross-check

Exomiser 14.0.0 on the same VCF ranks BUB1B first ($p = 0.0001$) with the same eight HPO terms; without phenotype input its top is a 46-way tie that still contains BUB1B. Both runs' record (`exomiser_summary.tsv` on the analysis host, artifact hashes verified 2026-09-17) are traceable; the answer is therefore not an artifact of this pipeline's own code.

## Reproduction

Python 3.12 with pysam, cyvcf2, pyyaml. References (paths in `pipeline/conf/track1_config.yml`): GENCODE v47 MANE, GRCh38 FASTA, gnomAD v4.1 exome sites, ClinVar, HPO with the OMIM gene-to-phenotype map. The CADD v1.7 option exists in the code but its tile set was not installed for this run, so all CADD fields in the ranked table are empty and the term is neutral.

```
python3 pipeline/scripts/prioritize.py \
    --conf pipeline/conf/track1_config.yml \
    --with-af \
    --with-hpo \
    --out output/annot_full.tsv
python3 pipeline/scripts/make_submission_csv.py \
    --in output/annot_full.tsv \
    --out submission_track1.csv
```

The annotate and HPO modules carry built-in self-tests (`python3 pipeline/scripts/annotate.py`, `python3 pipeline/scripts/_hpo.py`); they print SKIP with the missing reference named when a host lacks the reference files.

## Disclosure

Built with AI assistance (Kimi K3 via Oh My Pi); no patient data left the analysis hosts and none was sent to any external API, per the challenge rules.